



جامعة أم القرى
UMM AL-QURA UNIVERSITY

SYSTEM OF INSTITUTE

**Report Project object oriented programming
CS students
College of Computing
Department of Computer Science and Artificial Intelligence
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PROJECT DESCRIPTION:

Our idea is about a small institute for enrollment in one of the following colleges:

- Computer
- Management

We have also made some services available for professors.

FOR DETAILS :

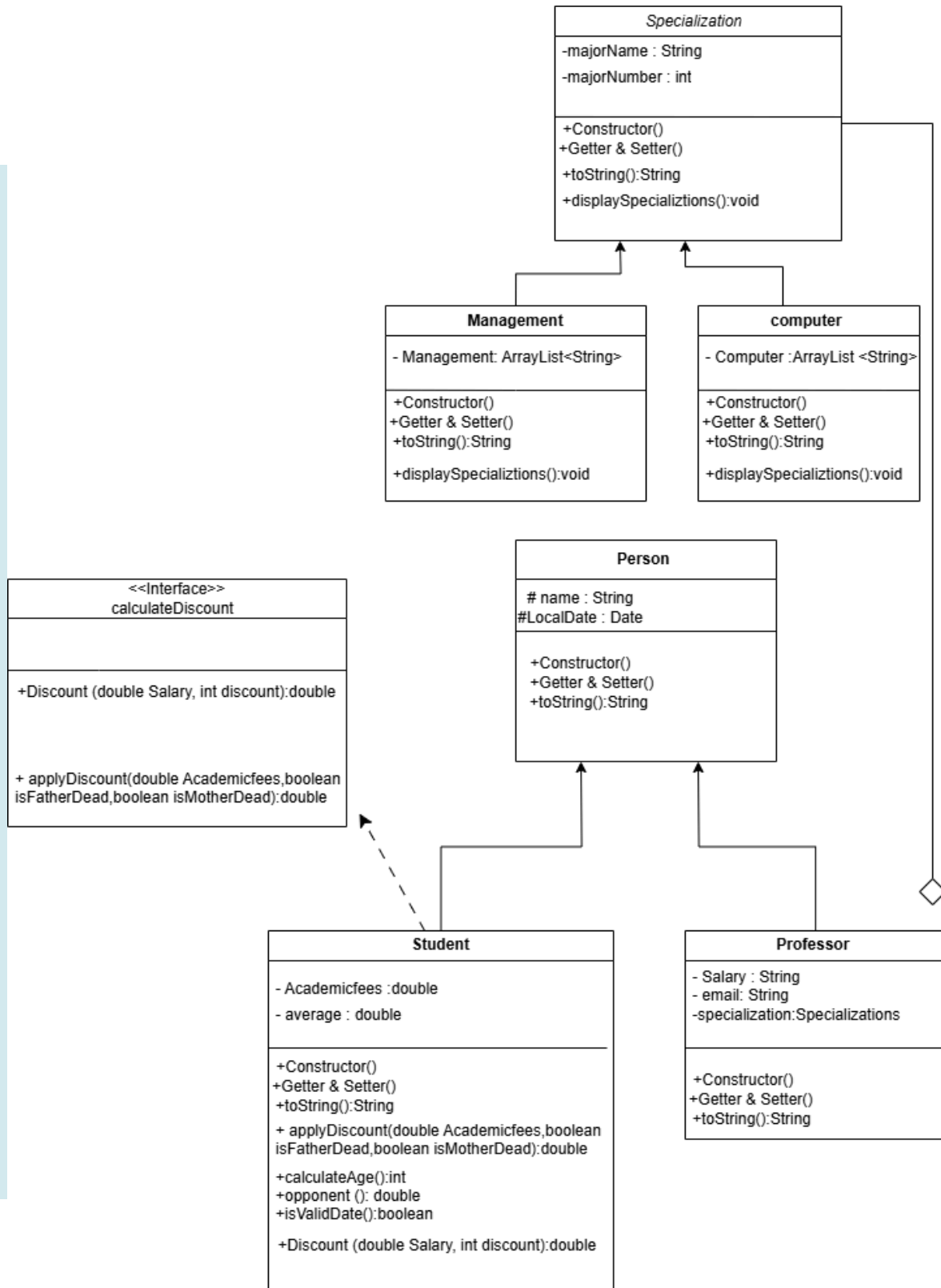
FOR STUDENTS:

1. Age Verification : We check if their age is appropriate for a university student.
2. College Choice :After verifying their age, we present them with two options: Computer college or management.
3. Specializations : Upon selecting a college, the available specializations are displayed. For example, if they choose Computer , they can see specializations like Computer Engineering.
4. Discount :If the student is an orphan, we offer a special discount. We first show them the tuition fee, then ask about their parents' status, as the discount varies based on the situation.

FOR PROFESSORS:

1. Identity Verification : We ask for their name, which will be recorded in the program.
2. Available Services : We provide them with the following options:
 - View salary.
 - View specialization.
 - View their email.
 - View all of this information.
 - Exit Option : Finally, they have the option to exit the program.

UML DIAGRAM FOR CALSSES :



OUTPUT :

```
run:
Welcome to the University!

Are you a Student or a Professor? (1 or 2 please enter just one Number):
1. Student
2. Professor
1
Please enter your name:
Duha
Enter birth year (YYYY):
2005
Enter birth month (MM):
5
Enter birth day (DD):
5
You are eligible to study. Your age is: 19
```

First, we welcome the person, then we ask if they are a student or a professor by entering the number (1 for student, 2 for professor).

We use an if-else statement.

If they choose the number 1, we present them with questions to enter their details (input their name and birth date).

We calculate their age using the `calculateAge` method, which uses the `Period` to compute the age based on the difference between the birth date and the current date.

also implemented a check for age to ensure that the allowed age for studying is between 18 and 25.

```
Please choose your Major (1 or 2 please enter just one Number):
1. Computer
2. Management
1
Computer specializations available :
1.Computer Science
2.Artificial Intelligence
3.Data Science
4.Web Development
5.Computer Engineering

Please choose your specialization (Please write the name of the Major's):
Computer Science
```

We give the user a choice between the two colleges, and they select a number (1 for Computer Science, 2 for Business Administration).

After that, the available specializations in the chosen college will be displayed, and we ask them to write the name of the specialization.

```
1
You must enter the name of the specialization, not a number.
Please choose your specialization (Please write the name of the Major's):
```



If they enter a number or leave it blank, the following message will be displayed:

OUTPUT :

```
Please choose your specialization (Please write the name of the Major's):
Computer Science

The university fees is: 15000.0

Is your father dead? (yes/no):
yes
Is your mother dead? (yes/no):
no
Final fee after discount: 13500.0

Well, here is all your information:
Your name is: Duha
Your age is: 19
Your Major is: Computer Science
Your university fees is: 13500.0
Your Professor Name is: D.Azhar
Your Professor Email is: azhar@gmail.com

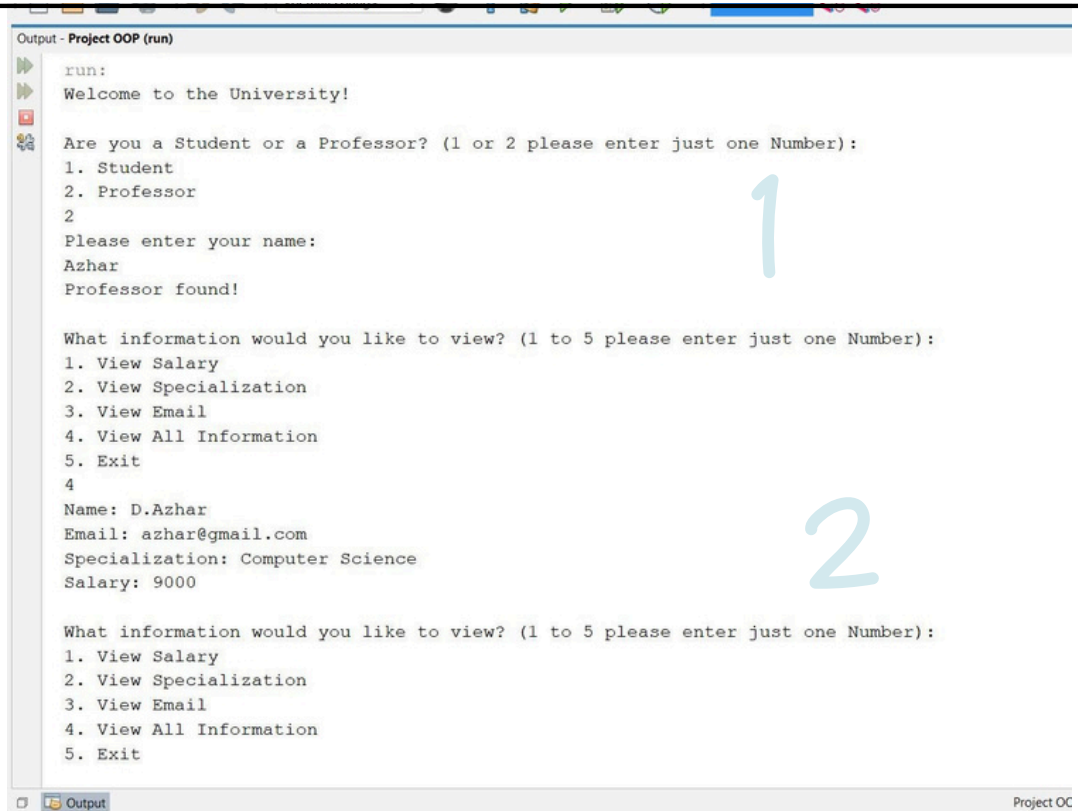
BUILD SUCCESSFUL (total time: 4 seconds)
```



When he finishes choosing his specialization, we display the fees. Then we ask him if his parents are alive. If he is an orphan, we apply a 10% discount. After that, we show him the amount after the discount.

At the end of the student services, we display all of his information and then present him with the professor associated with the specialization that he chose. The professors are stored in an array, and we retrieve their information along with the email of the corresponding professor using a `for` loop. We also added an `if-else` statement to check whether the specialization chosen by the student matches the specialization of the professor, and then we display all the information.

OUTPUT :



```
Output - Project OOP (run)
run:
Welcome to the University!
Are you a Student or a Professor? (1 or 2 please enter just one Number):
1. Student
2. Professor
2
Please enter your name:
Azhar
Professor found!

What information would you like to view? (1 to 5 please enter just one Number):
1. View Salary
2. View Specialization
3. View Email
4. View All Information
5. Exit
4
Name: D.Azhar
Email: azhar@gmail.com
Specialization: Computer Science
Salary: 9000

What information would you like to view? (1 to 5 please enter just one Number):
1. View Salary
2. View Specialization
3. View Email
4. View All Information
5. Exit
```

1 Just like we did with the student, we welcomed the professor and asked for his name. The name is stored in an array along with his specialization and salary. When displaying the professor's name, we ignored the capitalization and spaces, and added the prefix "D." before the professor's name. We also checked whether this name exists or not.

2 Here, we displayed the services offered to the professor. In this output, we chose, for example, to view all the information, so it printed his name, email, salary, and specialization. It will continue to show the services until he chooses to exit.

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THE ALLOCATION OF TASKS AND CLASSES :

Tasks	student name
Person	Jana
Professor	Duha
Student	Tala
Specializations	Jana
Computer	Tahani
Management	Tahani
calculateDiscount	Duha
Main	Sara, Tala



STUDENTS' NAMES:

TALA MAZEN FADEL	445007293
TAHANI AMER ALBARIQI	445006007
DUHA NASIR ALNABATI	445005974
SARA SULTAN ALMATRFI	445005027
JANA AHMAD ALMATRFI	445001958